

PHC Pulse — Demo Scenarios & Acceptance Tests

Filled · WHAT must actually work?

Purpose. Define what DONE means. The build must pass these scenarios, not just generate many screens.

Filled from the repository PHC-Pulse-GDG-Code-for-community (main @ fd42d41) on 26 Sep 2026. Numbers quoted were measured by running the engine and the 13-test pytest suite (all passing) with PHC_TODAY=2026-09-25, seed 42.

Status legend: **[IMPLEMENTED]** in the repo and tested **[PARTIAL]** partly built **[PLANNED]** designed here, not yet in code

1. Definition of Done

Global requirements for a complete build

- **Every S01–S06 scenario passes** end to end on seeded data, through the API and in the UI, with no manual database edits. S03 (expiry rescue) is the one open item.
- `cd backend && python -m pytest -q` passes: currently **13/13 green** in about 27 s (11 API/NLU tests plus 2 Gemini-path tests with a mocked model).
- The app runs with **and without** Gemini credentials. Every AI output shows its engine; there are no silent failures.
- Row-level scope is enforced on every read endpoint; officers can approve only in scope; PHC staff cannot approve.
- `npm run build` and `oxlint` are clean. The Docker image builds and `scripts/deploy.sh` deploys to Cloud Run; `/api/health` returns ok.
- The full engine recompute (120 PHCs × 10 drugs × 180 days) takes under 2 s (measured about 0.7 s). Pages render at 375 px width without horizontal scroll.

2. Critical Scenarios

ID	Scenario	Given	When	Then	Status / evidence
S01	Stockout	Low stock + rising demand	Forecast runs	Risk + depletion date shown	[x] PASS PHC-22 PCM: 174 strips, 21.3/day → high, 9 days, P = 1.0, next supply in 20 days
S02	Redistribution	Donor surplus + recipient risk	Optimizer runs	Donor + quantity + cost + ETA shown	[x] PASS REC-C4290993 PHC-14 → PHC-22, 284 strips, 41.5 km, INR 1,347, 3.2 h (test_deck_scenario)
S03	Expiry rescue	Surplus nearing expiry	Engine evaluates	Rescue recommendation shown	[] NOT YET Needs batch/expiry data and the rule in the AI spec §3
S04	Demand shock	Footfall spike	Anomaly runs	Alert + supply response shown	[x] PASS CL-KRP-diarrhoea (4 PHCs, 4.84×) and CL-GKP-fever; ORS/ZNC surge; IDSP draft (test_briefing_and_alert)
S05	Approval	Recommendation exists	Officer approves	Transfer created	[x] PASS status approved, approved_by, history; PHC role → 403 (test_transfer_lifecycle, test_phc_staff)
S06	Tracking	Transfer exists	Status updates	Dashboard reflects outcome	[x] PASS in_transit → delivered moves stock; delivered before in_transit → 400; forecasts recompute

3. API / Integration Tests

Test	Expected	Pass
Create report	POST /api/reports/parse (PHC-22, "Paracetamol 150 strips, ORS 60 sachets, beds occupied 3, staff present 9, OPD 72") → PCM 150, ORS 60, beds 3, staff 9, OPD 72; /reports/submit → PCM stock = 150. Also USSD *123# 5-step flow → "END Submitted"; SMS saved; Dialogflow reply in Hindi.	[x] PASS
Forecast	GET /api/overview → 120 PHCs, ≥ 2 clusters (diarrhoea, fever), backtest recall_14d_ahead > 0.6 (measured 0.814). Items expose days_to_stockout, p_stockout and status.	[x] PASS
Recommendation	GET /api/redistribution contains the lane PHC-14 → PHC-22 with PCM; each shipment has lines, distance_km, cost_inr, eta_hours, urgency_days.	[x] PASS
Approve transfer	POST /api/redistribution/{rec}/approve (national) → status approved; PHC-22 persona → 403 (tested). Stale id → 409 and out-of-scope → 403 are implemented but not yet covered by a test.	[~] PARTIAL

Test	Expected	Pass
Update status	delivered before in_transit → 400. in_transit then delivered → 200, recipient stock + qty.	[x] PASS
AI unavailable	No Gemini: voice upload → 503 with a "Gemini" message; text parse uses the rule parser; briefing and alert return template output tagged rules-fallback. With mocked Gemini: engine = model name and hallucinated drug code dropped.	[x] PASS

4. UI Tests

Navigation, responsive layout, forms, validation, loading, empty and error states

- **[~] PARTIAL** No automated UI tests in the repo yet. Checked by hand; add Playwright specs for the items below.
- **Navigation:** all 8 routes load from the sidebar and deep links (/phc/22). Persona switch shows a toast, reloads data and narrows the state/district filters.
- **Responsive:** at 375 × 812 the sidebar is a drawer opened by the menu button, KPI grid stacks, tables scroll inside their cards, and the page itself never scrolls sideways.
- **Forms/validation:** the voice recorder stops at 60 s; empty speech shows the hint; the SMS simulator shows the format hint for bad input; USSD accepts digits only.
- **Loading:** spinner labels ("Connecting to PHC Pulse...", "Solving the redistribution MILP..."). **Empty:** each list shows its empty message (see UI spec §4). **Error:** API errors show in ErrorBox; 403, 409 and 503 messages are readable.
- **Approval UX:** Approve → the card moves to the tracker; Dispatch / Delivered buttons appear only for legal next states.

5. Data / Model Tests

Exact synthetic scenario IDs and expected outputs (seed 42, PHC_TODAY=2026-09-25)

Scenario ID	Setup (seed.py)	Expected output
SC-PCM-22	PHC-22 paracetamol pinned to 8.6 days' cover, supplied 10 days ago	status high, days_to_stockout 9, next_supply_in 20, p_stockout ≥ 0.9
SC-PCM-14	PHC-14 over-indent paracetamol (bias 3.2), 140 days' cover	status surplus, surplus_units ≈ 3,038
SC-LANE-14-22	Both above, 41.5 road-km apart, same state	Shipment PHC-14 → PHC-22, PCM 284, INR 1,347, ETA 3.2 h, reason text as in the deck
SC-KRP-ADD	Diarrhoea ramp 1.8–5.2× on the 4 nearest Koraput PHCs	Cluster CL-KRP-diarrhoea = {PHC-91, 93, 95, 100}, severity high, max ratio ≈ 4.8
SC-GKP-FEV	Fever ramp 1.6–3.0× on 3 Gorakhpur PHCs	Cluster CL-GKP-fever = {PHC-61, 62, 66}, max ratio ≈ 2.9
SC-KLB-RESP	Respiratory 2.6–3.1× at one Kalaburagi PHC	PHC-51 anomaly, in_cluster = false
SC-BACKTEST	180-day history	recall_14d_ahead 0.814, precision 0.671, weekly_accuracy 0.79 (± 0.02 regression band)
SC-SCOPE	dho-TVM token, ?state=UP	15 PHCs, all TVM (scope cannot widen)

6. Demo Script

Setup → worker input → forecast → optimizer → approval → outcome

- **Setup (0:00):** deploy or run locally with PHC_TODAY=2026-09-25 and a Gemini key; POST /api/admin/reset. Open the **Command centre** as national: 120 PHCs, predicted stock-outs, two clusters on the map. Generate the AI briefing and switch it to Hindi.
- **Outbreak (0:45): Outbreak signals** → Koraput diarrhoea cluster (4 PHCs, up to 4.8×). Draft the IDSP alert (English + Odia); show ORS/zinc forecasts raised automatically.
- **Worker input (1:30): Voice report** for PHC-22 in Tamil, about 20 s. Show the transcript, translation, fields and the Tamil read-back, answer the follow-up, and Submit. Then the feature-phone page: *123#, an SMS report and a Hindi IVR call.
- **Forecast (2:30):** open PHC-22: paracetamol runs out in 9 days, next supply in 20, with the forecast chart and interval.
- **Optimizer + approval (3:00):** switch to **DHO Tiruvannamalai** (scope narrows to 15 PHCs). On **Redistribution**: "PHC-14 has surplus paracetamol; PHC-22 needs it in 9 days", 284 strips, INR 1,347, 3.2 h. Approve → Dispatch → Delivered.

- **Outcome (3:45):** PHC-22 stock rises and its status clears; the tracker history shows who did what. Close on **Models & HMIS:** 81% of stock-outs flagged ≥ 2 weeks ahead, HMIS CSV export, and the BigQuery / Vertex path.

7. Judge Questions

Question	Prepared answer / evidence
What is novel?	It closes the loop from voice to delivery: a one-minute local-language report → forecast that only alerts when the stock-out beats the next indent → outbreak-aware demand → a MILP that consolidates lanes and weights drug criticality → one-tap approval that moves the ledger. It works on a feature phone.
Why AI?	Gemini removes the form: multilingual, code-switched speech to validated JSON in one call. Forecasting and anomaly models see trouble weeks ahead where monthly HMIS does not. The optimiser picks among thousands of donor/recipient/drug combinations. Gemini briefings turn numbers into actions.
Why not existing HMIS?	HMIS is monthly, backward-looking and aggregate, with no forecast, no facility-to-facility transfer and no outbreak footfall signal. PHC Pulse keys on the HMIS NIN and exports HMIS-style CSV: it complements HMIS and does not replace it.
What data is real?	Real: district centroids, block names, NLEM drug list, HMIS-like PHC OPD ranges, state epidemiology patterns. Synthetic: all daily stock and footfall, PHC coordinates (jittered), outbreaks (3 injected). Labelled in the UI. No patient data.
How is accuracy measured?	Leakage-free walk-forward backtest on the last 60 days: recall 81.4% of stock-outs flagged ≥ 14 days ahead, precision 67.1%, weekly wMAPE 0.21. Anomaly detection checked against injected outbreaks: 2/2 clusters, 8/8 PHCs. Pilot plan: compare with real HMIS stock-out days.
What if AI is wrong?	Nothing moves without an officer. Voice numbers are read back and confirmed. Drug codes are whitelisted and every output shows its engine. Rule-based fallbacks keep it running if Gemini fails. Donors keep safety stock (1.2× forecast + 7 days), so a wrong transfer cannot empty them.
How does it scale?	The MILP separates by state (one small problem each; national is about 1,100 variables, solved in seconds). Forecasting is vectorised (1,200 series in ms), then moves to BQML ARIMA_PLUS / Vertex. BigQuery is partitioned by day and clustered by state, with row policies. Cloud Run is stateless once the store moves off SQLite. Roadmap: 100-PHC pilot → state rollout → cross-state → blood, oxygen, ambulances.